

TURNING RED



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Measuring Antibiotic Resistance with Color

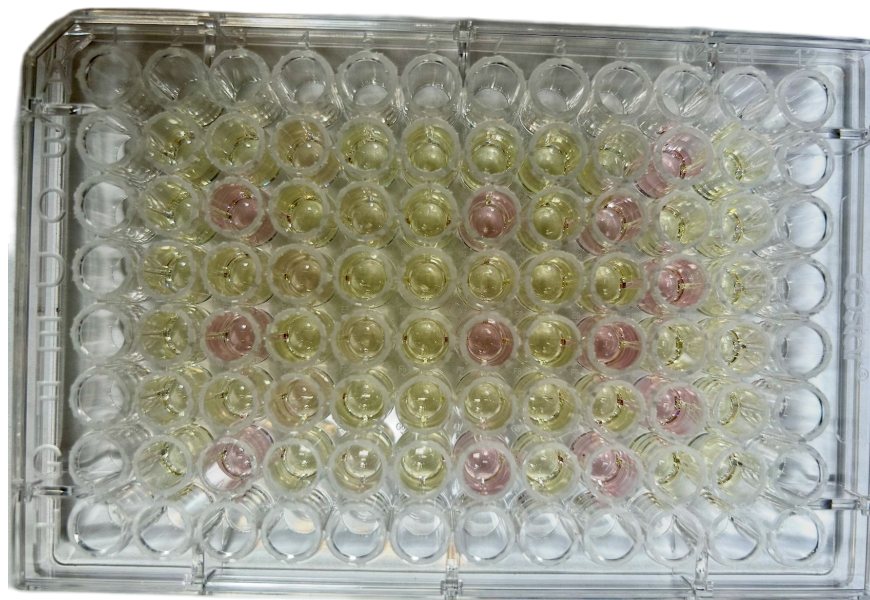
We developed a robotic workflow to test many compounds for their ability to suppress antibiotic resistance.

● → ● Antibiotics will work! ✓

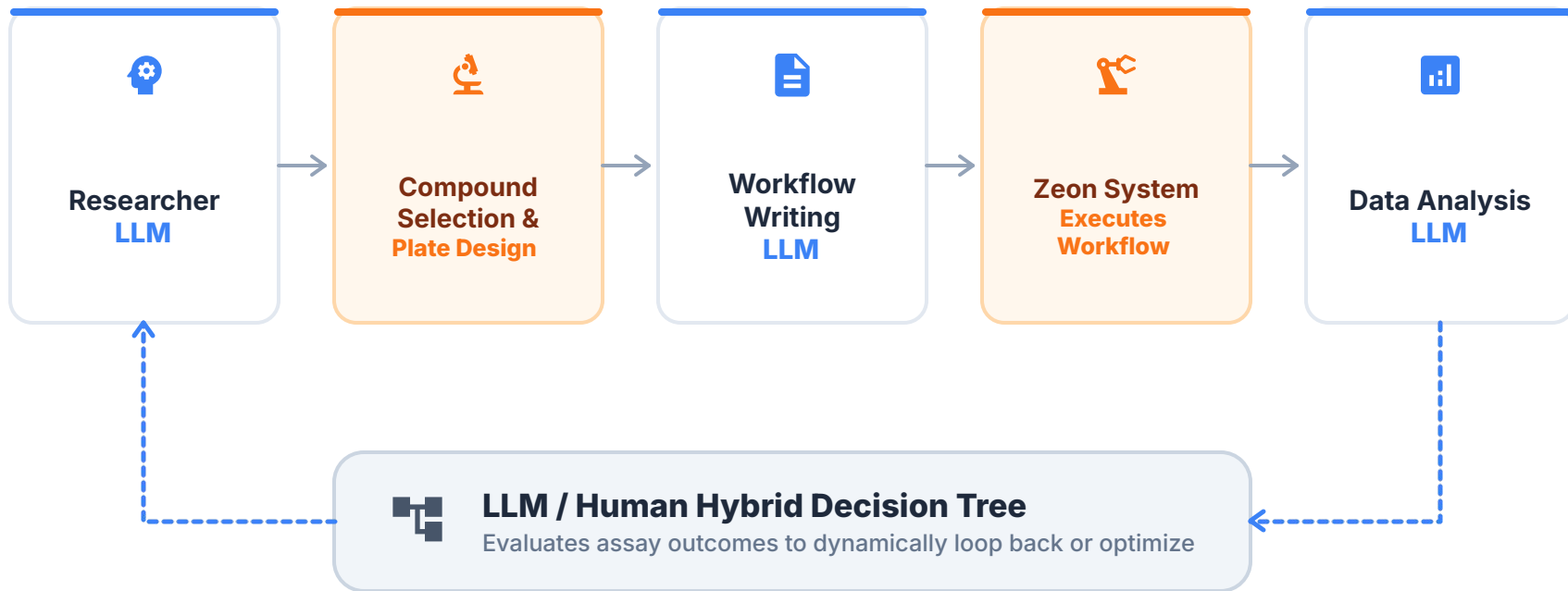
● → ● Antibiotic resistance !

Nitrocefin Readout

We use Nitrocefin as a colored readout for rapid visual detection of antibiotic resistance.

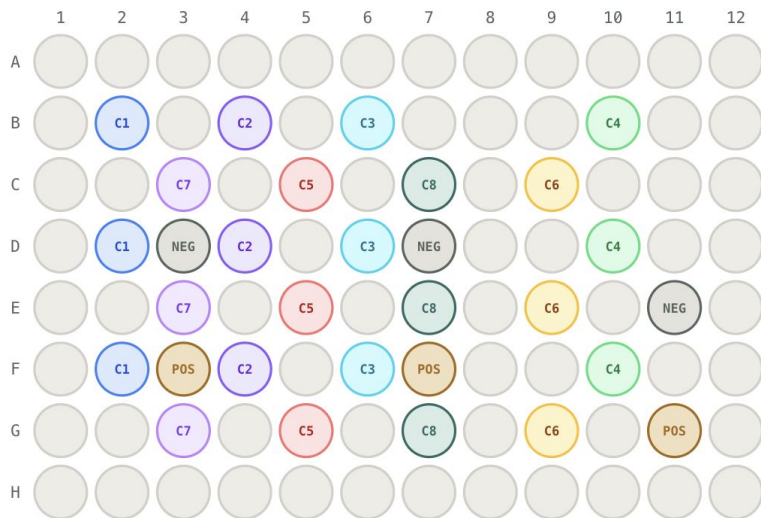


AI Agents Close the Loop for Scientific Assay Development

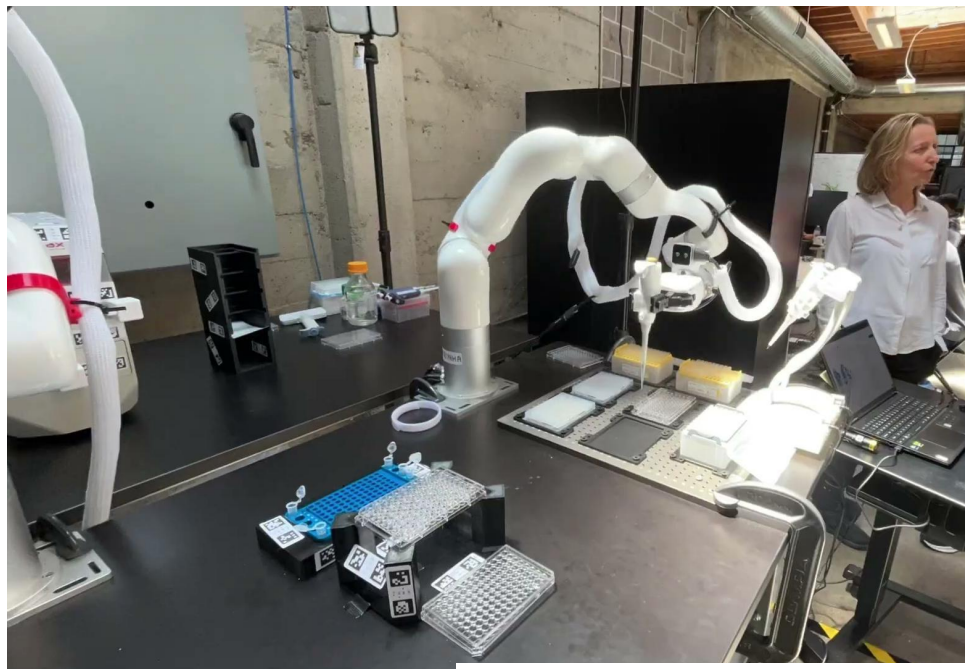


Flexible Robot (Random Access Pipetting)

Custom plate layout mapping for automated multi-compound testing workflow



● Positive ctrl (T1262) ● Negative ctrl (no enzyme) ● C1 T1262 Tazobactam ● C2 T6685 Sulbactam ● C3 T14081 Enmetazobactam
● C4 T1008 Cephalexin ● C5 T0224 Meropenem ● C6 T0985 Oxacillin ● C7 T0138 Cefpiramide ● C8 T8390 Cefazolin ● Empty



(time lapse, not real time)

Round 1: Baseline E2E Operation Timeline

Detailed sequence mapping of the robotic process prior to optimization

Baseline Profile

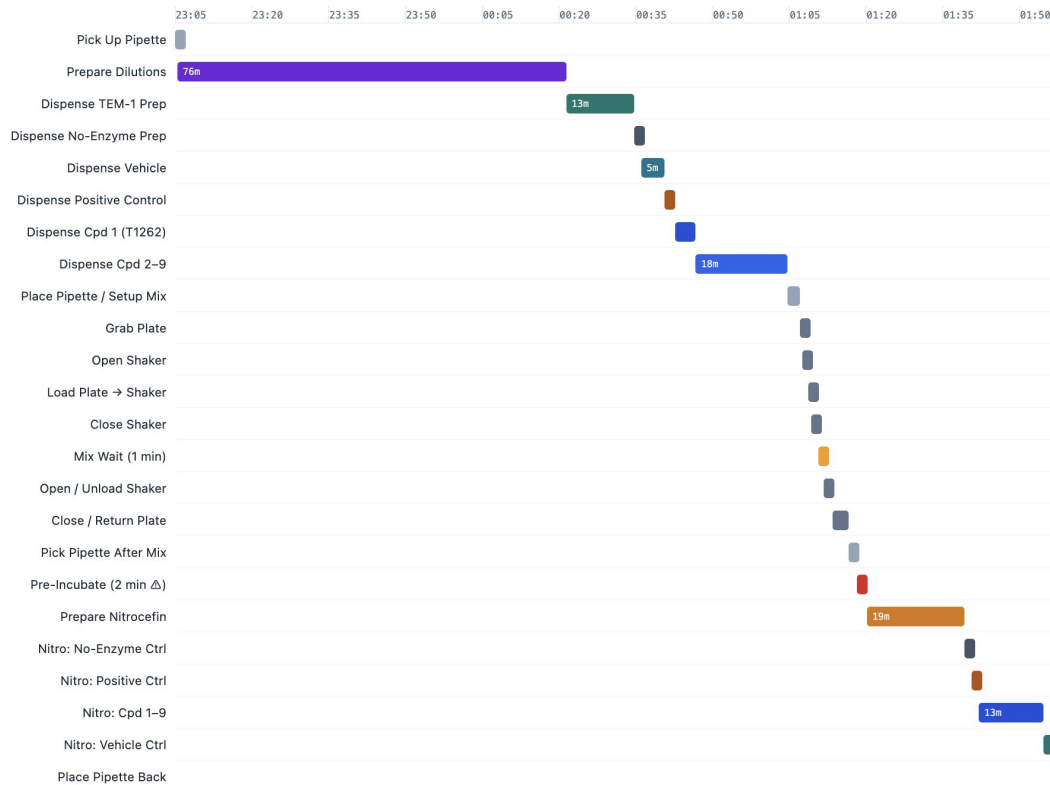
2h 45m

Total Run Time

Key Observations:

- Dilutions & compound prep account for major run time
- Robotic nitrocefin prep & dispensing delays affect readout accuracy (hinged on rapid TEM-1 cleavage kinetics)

Target: Reduce to under 1 hour and prep Nitrocefin fast



Round 2: Optimized E2E Operation Timeline

Detailed sequence mapping of the robotic process after optimization

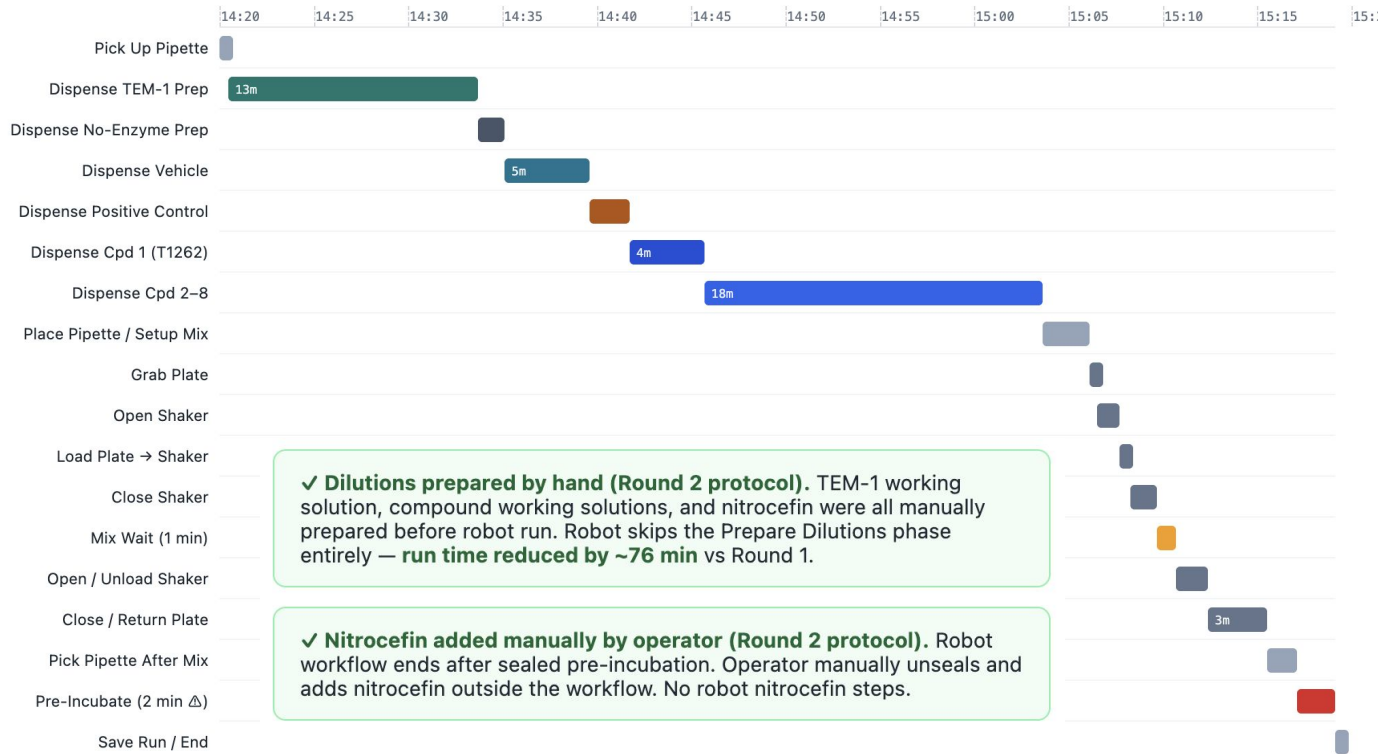
AI ACCELERATION

Researcher Agent accelerated the protocol by 2 hours

OPTIMIZED PROFILE

1h

Total Run Time



Data quality improved from round 1 to round 2

Round 1

Inconclusive and inconsistent

COMPOUND	% INHIB (MEDIAN)	LABEL
T0224 (meropenem @ 50 µM)	97.0%	surprise hit
T1005 (amoxicillin @ 50 µM)	68.2%	surprise hit
T6685 (sulbactam sodium @ 50 µM)	27.3%	surprise miss
T0138 (cefpiramide acid @ 50 µM)	—	inconclusive
T0985 (oxacillin sodium salt @ 50 µM)	—	inconclusive
T1008 (cephalexin @ 50 µM)	—	inconclusive
T1262 (tazobactam @ 1 µM)	—	inconclusive
T14081 (enmetazobactam @ 50 µM)	—	inconclusive
T8390 (cefazolin @ 50 µM)	—	inconclusive

3 callable · 6 inconclusive on endpoint median

(22 wells failed 0–150% window · 16 min nitrocefin stagger)

Round 2

Controls worked, we found a potential hit

COMPOUND	% INHIB (MEDIAN)	LABEL
T1262 (tazobactam @ 1 µM)	106.9%	confirmed hit
T14081 (enmetazobactam @ 50 µM)	97.2%	confirmed hit
T0224 (meropenem @ 50 µM)	87.5%	surprise hit
T6685 (sulbactam sodium @ 50 µM)	58.3%	confirmed hit
T0985 (oxacillin sodium salt @ 50 µM)	19.4%	confirmed substrate
T1008 (cephalexin @ 50 µM)	0.0%	confirmed substrate

6 discovery compounds on endpoint median

Q2E/Q3 PASS · sync dose ~2.0 min

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